

MATHILDE KAPPEL

PhD Student (3rd Year), Sorbonne University, Paris, France ◊ kappel@isir.upmc.fr

DEGREES

PhD in Learning Robotic Manipulation

Sept. 2024 – Sept. 2027

Sorbonne University – Institute of Intelligent Systems and Robotics (ISIR), Paris, France

Thesis: From Data Generation to Skill Extraction in Robotics – Learning Grasping and Object Manipulation Behaviors.

Advisors: M. Khoramshahi, L. Annabi, F. Ben Amar, S. Doncieux

Expected completion: end of 2027.

Master's Degree in Robotics and Automation

Sept. 2022 – Sept. 2023

Clermont Auvergne University, Clermont-Ferrand, France

Specialization: Artificial Perception and Robotics – Dual-degree program.

Master's Degree in Mechanical Engineering

Sept. 2020 – Sept. 2024

SIGMA Clermont Engineering School, Clermont-Ferrand, France

Specialization: Robotics and Innovative Mechanisms. Ranked among the top 10 students in the program.

Two-Year Intensive Preparatory Program

Sept. 2018 – Sept. 2020

Bellevue High School, Toulouse, France

Preparation for competitive entrance examinations to engineering schools.

Scientific Baccalaureate – Highest Honours

Sept. 2015 – Sept. 2018

Ozenne High School, Toulouse, France

EXPERIENCE

Research in Robotic Grasping

Sept. 2023 – Sept. 2024

Sorbonne University – Institute of Intelligent Systems and Robotics (ISIR), Paris, France

Research on robotic grasping and manipulation (internship and employment contract).

Supervised by: Stéphane Doncieux, Professor of Computer Science.

Research Intern in Minimally Invasive Robotics

March – Sept. 2023

Jesús Usón Minimally Invasive Surgery Centre (CCMIJU), Cáceres, Spain

Supervised by: Juan A. Sánchez Margallo, Head of Robotics, Image-Guided Surgery, and Bioprinting.

Research Intern in Gait Estimation for Walking Exoskeletons

May – Sept. 2022

InteSpring Exoskeletons, Rijswijk, Netherlands

Supervised by: Trebsijg vd Wijdeven, Electronics, Technology, and Project Management.

Production Operator Intern in Electronics Assembly Department

July – Aug. 2021

Thales Alenia Space, Toulouse, France

Supervised by: Sandrine Lelarges, Production Unit Manager.

PUBLICATIONS

Papers at Refereed Conferences

- **QDTraj: Exploration of Diverse Trajectory Primitives for Articulated Object Robotic Manipulation.** M. Kappel, M. Khoramshahi, L. Annabi, F. Ben Amar, S. Doncieux.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

- **QDGset: A Large-Scale Grasping Dataset Generated with Quality-Diversity.** J. Huber*, F. H      *, M. Kappel*, I. de Loyola P        *, S. T. Puente, P. Gil, F. Ben Amar, S. Doncieux.
IEEE International Conference on Robotics and Automation (ICRA), 2025.
- **Speeding up 6-DoF Grasp Sampling with Quality-Diversity (QDG-6DoF).** J. Huber*, F. H      *, M. Kappel*, E. Chelly, M. Khoramshahi, F. Ben Amar, S. Doncieux.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.

Workshop and Conference Papers

- **A Sim-to-Real Integration Pipeline for Training and Deployment of Chunk-Based VLA Manipulation Policies.** M. Kappel*, C. Grislain*, M. Chetouani, O. Sigaud, L. Annabi, F. Ben Amar, S. Doncieux, M. Khoramshahi.
IROS 2026 Workshop on Sim2Real and Classical Control, 2026.
- **ROS2 Architecture for Articulated Object Manipulation Using Trajectory Primitives Learned in Simulation.** M. Kappel, M. Khoramshahi, L. Annabi, F. Ben Amar, S. Doncieux.
ROSConFrance, 2026.
- **Towards Robust Prehensile Manipulation in Open-Ended Environments.** M. Kappel, M. Khoramshahi, L. Annabi, F. Ben Amar, S. Doncieux.
International Conference on Intrinsically Motivated Open-ended Learning (IMOL), 2025.

Other Publications

- **Application of Artificial Intelligence for Stress Prediction in Minimally Invasive Robotic Surgery.** D. Caballero, J. A. S            , M. J. P            , M. Kappel, F. M. S            .
XLI Annual Congress of the Spanish Society of Biomedical Engineering (CASEIB), 2023.
- **Use of Electrodermal Activity (EDA) for the Classification of Stress Situations in Minimally Invasive Robotic Surgery.** D. Caballero, J. A. S            , M. J. P            , F. M. S            , M. Kappel, F. M. S            .
XLI Annual Congress of the Spanish Society of Biomedical Engineering (CASEIB), 2023.

* Equal contribution.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Sorbonne University

2024 – Present

Computer science teaching at undergraduate and graduate levels.

- Bachelor’s Year 1: Programming Fundamentals, Practical classes, 66.8h; Data Science, Practical classes, 38.5h;
- Bachelor’s Year 2: Computer Science and Statistics, Practical classes, 12h; Underwater Observation Robot Design, Project supervision, 53.5h;
- Master’s Year 1: Robotics and Machine Learning, Practical classes, 12h; Android Project, Project supervision, 2h.

TECHNICAL SKILLS

Robotics: Robot Manipulation, Trajectory Learning, Evolutionary Algorithms, Quality-Diversity, Robot Control, Sim-to-Real, Franka FR3 Real Robot.

Programming: Python, C++, **Libraries:** PyTorch, OpenCV, Open3D, Pixi, **Sim Engine:** Genesis, Pybullet, **Software:** Onshape, 3DEXPERIENCE, SolidWorks, MATLAB, MeshLab **Real Robot:** ROS2, MoveIt2, **Computing:** Ubuntu, Singularity, GPU, slurm.

Languages: French (Native), English, Spanish.

ACADEMIC ACTIVITIES

- **ROMANDIC Winter School on Robot Handling of Deformable Objects** *Feb. 2026*
Kranjska Gora, Slovenia.
Training and hands-on sessions on robotic manipulation of deformable objects, with a focus on perception, simulation, and robot manipulation.
- **Participation in Robotics Coopetition** *2024*
Paris, France.
Participation in the euROBIN project and the Hugging Face Hackathon.
- **Final-Year Engineering Projects** *2023*
Clermont-Ferrand, France.
ROS Bookstore Inventory Robot: developed a ROS-based mobile robot for automated stock inventory in a bookstore; Isoglide Parallel Robot: designed and modeled a four-arm parallel robot using CAD, including kinematic part design, 3D assembly, mechanism simulation, and control.

IEE Conference Attendance

- **IROS 2026 – International Conference on Intelligent Robots and Systems** *2026*
Pittsburgh, USA.
- **ICRA 2025 – International Conference on Robotics and Automation** *2025*
Atlanta, USA.
- **IROS 2024 – International Conference on Intelligent Robots and Systems** *2024*
Abu Dhabi, UAE.

PROFESSIONAL AND COMMUNITY ENGAGEMENT

- **Mentor and Ambassador, Elles Bougent Association**
Supporting initiatives to promote engineering and scientific careers among young women.
- **Student Member, IEEE Robotics and Automation Society (RAS)**

OPEN-SOURCE PROJECTS

- **QDTraj – Diverse Trajectory Primitives for Articulated Object Manipulation**
Source code: https://gitlab.isir.upmc.fr/kappel/qdtraj_public_source_code
Project webpage: https://kappel.web.isir.upmc.fr/trajectory_primitive_website/
CAD model: [Onshape-cad.hyperlink](#)
Dataset: https://huggingface.co/datasets/mathildekappel/trajectory_primitive
- **Sim-to-Real Chunk-Based Control**
Source code: https://gitlab.isir.upmc.fr/kappel/sim2real_public_chunk_control/
- **QDGset – Large-Scale Grasping Dataset**
Dataset: https://huggingface.co/datasets/QDGrasp/QDGset_no_shapenet
- **QDG-6DoF – 6-DoF Grasp Sampling with Quality-Diversity** (partial contribution)
Source code: https://gitlab.isir.upmc.fr/l2g/qd_grasp_6dof
Project webpage: <https://qdgrasp.github.io/>

ONLINE PROFILES

Google Scholar: <https://scholar.google.com/citations?user=8xfzmUYAAAAJ&hl=fr&oi=ao>

LinkedIn: <https://www.linkedin.com/in/mathilde-kappel-528bb11b9/>

Research YouTube Channel: <https://www.youtube.com/@MathildeKappelResearch>